

Chapter 4: Fractions and percentages

12 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems.

CHAPTER

4

4.01 Fractions

4.02 Ordering fractions

4.03 Adding and subtracting fractions

4.04 Adding and subtracting mixed numerals

4.05 Fraction of a quantity

4.06 Multiplying fractions

4.07 Dividing fractions

4.08 Fraction problems

4.09 Fractions on a calculator

4.10 Percentages, fractions and decimals

4.11 Percentage of a quantity

4.12 Expressing as a fraction or percentage

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

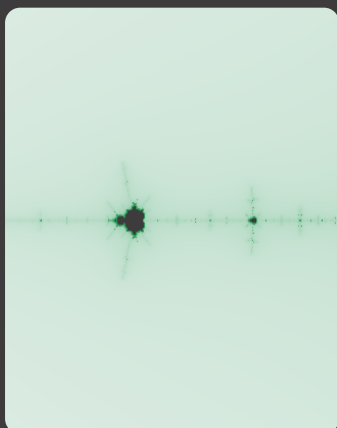
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

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In this chapter you will

- write equivalent fractions and simplify
- order fractions
- add, subtract, multiply and divide fractions and mixed numerals
- find a fraction or percentage of a quantity
- solve fraction problems
- convert between fractions, decimals and percentages



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
4.01

Fractions

Date _____

Learning intentions

We are learning to:

- convert between mixed numerals and improper fractions
- write equivalent fractions and simplify fractions

Success criteria

I can:

- ☐ change a mixed numeral to an improper fraction and back
- ☐ multiply top and bottom by the same number to find an equivalent fraction
- ☐ divide top and bottom by the HCF to simplify

Key terms Write the definition as your teacher explains it.

Improper fraction

Mixed numeral

Equivalent

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Convert between mixed numerals and improper fractions.

a Write $1\frac{3}{4}$ as an improper fraction.

b Write $\frac{7}{4}$ as a mixed numeral.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write each mixed numeral as an improper fraction.

12 QUESTIONS

REMEMBER Mixed → improper: whole × bottom + top.

1 $1\frac{3}{4}$

2 $2\frac{4}{9}$

3 $5\frac{3}{5}$

4 $3\frac{1}{7}$

5 $4\frac{1}{2}$

6 $2\frac{5}{6}$

Questions 7–12: simplify each fraction.

7 $\frac{6}{8}$

8 $\frac{10}{15}$

9 $\frac{12}{16}$

10 $\frac{9}{27}$

11 $\frac{14}{21}$

12 $\frac{20}{50}$

YOU DO

On your own: **Easy** Set B. Write each improper fraction as a mixed numeral.

6 QUESTIONS

① $\frac{7}{4}$

④ $\frac{11}{2}$

2 $\frac{26}{3}$

5 $\frac{17}{5}$

3 $\frac{23}{7}$

6 $\frac{36}{24}$

 Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Complete each pair of equivalent fractions.

a Complete: $\frac{1}{3} = \frac{?}{15}$

Your teacher works this out. Copy every step.

b Complete: $\frac{3}{4} = \frac{12}{?}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Complete each pair of equivalent fractions.

6 QUESTIONS

REMEMBER Improper $\rightarrow$ mixed: divide top by bottom.

① $\frac{1}{3} = \frac{?}{15}$

[illegible]

3 $\frac{4}{5} = \frac{?}{20}$

Case No.	Case Name	Case Address	Case City	Case State	Case Zip	Case Phone	Case Email	Case Date	Case Time	Case Status	Case Notes
1	John Doe	123 Main St	New York	NY	10001	212-555-1234	john.doe@example.com	2023-10-27	14:30	Completed	Initial assessment and data collection.
2	Jane Smith	456 Elm St	Los Angeles	CA	90001	310-555-5678	jane.smith@example.com	2023-10-28	10:00	In Progress	Interview with subject and review of records.
3	Robert Johnson	789 Oak St	Chicago	IL	60601	312-555-9012	robert.johnson@example.com	2023-10-29	09:00	Pending	Awaiting approval for field visit.
4	Maria Garcia	101 Pine St	San Francisco	CA	94101	415-555-3456	maria.garcia@example.com	2023-10-30	11:00	Completed	Final report and analysis completed.
5	David Lee	202 Birch St	Seattle	WA	98101	206-555-7890	david.lee@example.com	2023-10-31	13:00	In Progress	Interview with subject and review of records.
6	Emily White	303 Cedar St	Portland	OR	97201	503-555-2345	emily.white@example.com	2023-11-01	10:00	Pending	Awaiting approval for field visit.
7	Michael Brown	404 Maple St	Denver	CO	80201	303-555-6789	michael.brown@example.com	2023-11-02	14:00	Completed	Final report and analysis completed.
8	Sarah Davis	505 Spruce St	Phoenix	AZ	85001	602-555-0123	sarah.davis@example.com	2023-11-03	09:00	In Progress	Interview with subject and review of records.
9	James Wilson	606 Elm St	San Diego	CA	92101	619-555-4567	james.wilson@example.com	2023-11-04	11:00	Pending	Awaiting approval for field visit.
10	Lisa Anderson	707 Oak St	San Jose	CA	95101	408-555-8901	lisa.anderson@example.com	2023-11-05	13:00	Completed	Final report and analysis completed.
11	Kevin Taylor	808 Pine St	San Antonio	TX	78201	214-555-2345	kevin.taylor@example.com	2023-11-06	10:00	In Progress	Interview with subject and review of records.
12	Amanda Miller	909 Cedar St	San Jose	CA	95101	408-555-6789	amanda.miller@example.com	2023-11-07	14:00	Pending	Awaiting approval for field visit.
13	Christopher Hall	1010 Maple St	San Jose	CA	95101	408-555-0123	christopher.hall@example.com	2023-11-08	09:00	Completed	Final report and analysis completed.
14	Michelle King	1111 Birch St	San Jose	CA	95101	408-555-4567	michelle.king@example.com	2023-11-09	11:00	In Progress	Interview with subject and review of records.
15	Daniel Scott	1212 Spruce St	San Jose	CA	95101	408-555-8901	daniel.scott@example.com	2023-11-10	13:00	Pending	Awaiting approval for field visit.
16	Stephanie Adams	1313 Elm St	San Jose	CA	95101	408-555-2345	stephanie.adams@example.com	2023-11-11	10:00	Completed	Final report and analysis completed.
17	Gregory Baker	1414 Oak St	San Jose	CA	95101	408-555-6789	gregory.baker@example.com	2023-11-12	14:00	In Progress	Interview with subject and review of records.
18	Heather Clark	1515 Pine St	San Jose	CA	95101	408-555-0123	heather.clark@example.com	2023-11-13	09:00	Pending	Awaiting approval for field visit.
19	Timothy Evans	1616 Cedar St	San Jose	CA	95101	408-555-4567	timothy.evans@example.com	2023-11-14	11:00	Completed	Final report and analysis completed.
20	Christina Harris	1717 Maple St	San Jose	CA	95101	408-555-8901	christina.harris@example.com	2023-11-15	13:00	In Progress	Interview with subject and review of records.
21	Jonathan Lewis	1818 Birch St	San Jose	CA	95101	408-555-2345	jonathan.lewis@example.com	2023-11-16	10:00	Pending	Awaiting approval for field visit.
22	Karen Walker	1919 Spruce St	San Jose	CA	95101	408-555-6789	karen.walker@example.com	2023-11-17	14:00	Completed	Final report and analysis completed.
23	Steven Young	2020 Elm St	San Jose	CA	95101	408-555-0123	steven.young@example.com	2023-11-18	09:00	In Progress	Interview with subject and review of records.
24	Michelle King	2121 Oak St	San Jose	CA	95101	408-555-4567	michelle.king@example.com	2023-11-19	11:00	Pending	Awaiting approval for field visit.
25	Christopher Hall	2222 Pine St	San Jose	CA	95101	408-555-8901	christopher.hall@example.com	2023-11-20	13:00	Completed	Final report and analysis completed.
26	Michelle King	2323 Cedar St	San Jose	CA	95101	408-555-2345	michelle.king@example.com	2023-11-21	10:00	In Progress	Interview with subject and review of records.
27	Christopher Hall	2424 Maple St	San Jose	CA	95101	408-555-6789	christopher.hall@example.com	2023-11-22	14:00	Pending	Awaiting approval for field visit.
28	Michelle King	2525 Birch St	San Jose	CA	95101	408-555-0123	michelle.king@example.com	2023-11-23	09:00	Completed	Final report and analysis completed.
29	Christopher Hall	2626 Spruce St	San Jose	CA	95101	408-555-4567	christopher.hall@example.com	2023-11-24	11:00	In Progress	Interview with subject and review of records.
30	Michelle King	2727 Elm St	San Jose	CA	95101	408-555-8901	michelle.king@example.com	2023-11-25	13:00	Pending	Awaiting approval for field visit.
31	Christopher Hall	2828 Oak St	San Jose	CA	95101	408-555-2345	christopher.hall@example.com	2023-11-26	10:00	Completed	Final report and analysis completed.
32	Michelle King	2929 Pine St	San Jose	CA	95101	408-555-6789	michelle.king@example.com	2023-11-27	14:00	In Progress	Interview with subject and review of records.
33	Christopher Hall	3030 Cedar St	San Jose	CA	95101	408-555-0123	christopher.hall@example.com	2023-11-28	09:0		

5 $\frac{1}{2} = \frac{?}{50}$

② $\frac{3}{4} = \frac{12}{?}$

4 $\frac{5}{12} = \frac{15}{?}$

[illegible]

6 $\frac{3}{20} = \frac{?}{100}$

[illegible]

Level 3 · Challenge

Challenging · 6 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Simplify each fraction.

a Simplify $\frac{25}{30}$.

b Simplify $\frac{36}{24}$ and write as a mixed numeral.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Simplify each fraction.

6 QUESTIONS

REMEMBER Equivalent: $\times$ or $\div$ top and bottom by the same number.

1 $\frac{25}{30}$

3 $\frac{15}{20}$

5 $\frac{24}{48}$

2 $\frac{30}{40}$

4 $\frac{20}{15}$

6 $\frac{24}{42}$

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write $3\frac{2}{5}$ as an improper fraction.

Level 2

Complete: $\frac{2}{7} = \frac{?}{35}$

Level 3

Simplify $\frac{45}{60}$.

Extension · Stretch yourself**Worked example**

Find three fractions equivalent to $\frac{2}{3}$ with denominators between 10 and 20.

1. Multiply top and bottom by the same number
2. $\times 4$, $\times 5$, $\times 6$

Answer: $\frac{8}{12}$, $\frac{10}{15}$, $\frac{12}{18}$

YOU DO

On your own: read the worked example, then try these.

- E1 Which fraction is not equivalent to $\frac{3}{5}$: $\frac{9}{15}$, $\frac{12}{20}$, $\frac{15}{30}$, $\frac{30}{50}$?

- E2 A fraction simplifies to $\frac{2}{7}$. Its numerator and denominator add to 45. Find the fraction.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Mixed $\rightarrow$ improper: whole $\times$ bottom + top.
 2. Improper $\rightarrow$ mixed: divide top by bottom.
 3. Equivalent: $\times$ or $\div$ top and bottom by the same number.
 4. Simplify: $\div$ top and bottom by the HCF.

Level 1

$$1\frac{3}{4}$$

1. $1 \times 4 + 3 = 7$

Answer: $\frac{7}{4}$

Level 2

$$\frac{26}{3}$$

1. $26 \div 3 = 8 \text{ r } 2$

Answer: $8\frac{2}{3}$

Level 3

$$\text{Simplify } \frac{24}{42}$$

1. HCF is 6

Answer: $\frac{4}{7}$

Extension

$$\text{Equivalent to } \frac{2}{3}$$

1. $\times 4$, $\times 5$, $\times 6$

Answer: $\frac{8}{12}$, $\frac{10}{15}$, $\frac{12}{18}$

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LESSON
4.02

Ordering fractions

Date _____

Learning intentions

We are learning to:

- compare and order fractions

Success criteria

I can:

- ☐ compare fractions with the same denominator
- ☐ change fractions to a common denominator
- ☐ order fractions from smallest to largest

Key terms Write the definition as your teacher explains it.

Common denominator

Ascending order

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Which fraction is larger?

a Which is larger: $\frac{3}{8}$ or $\frac{5}{8}$?

b Which is larger: $\frac{2}{3}$ or $\frac{3}{5}$?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Which fraction is larger?

12 QUESTIONS

REMEMBER Same denominator: compare the numerators.

1 $\frac{3}{7}$ or $\frac{5}{7}$

2 $\frac{1}{4}$ or $\frac{1}{3}$

3 $\frac{9}{10}$ or $\frac{7}{10}$

4 $\frac{1}{2}$ or $\frac{2}{5}$

5 $\frac{2}{3}$ or $\frac{5}{6}$

6 $\frac{3}{4}$ or $\frac{5}{8}$

Questions 7–12: which is larger?

7 $\frac{2}{5}$ or $\frac{3}{5}$

8 $\frac{1}{2}$ or $\frac{3}{8}$

9 $\frac{2}{3}$ or $\frac{3}{4}$

10 $\frac{5}{6}$ or $\frac{7}{9}$

11 $\frac{3}{10}$ or $\frac{1}{4}$

12 $\frac{4}{7}$ or $\frac{1}{2}$

YOU DO

On your own: **Easy** Set B. Write each fraction with the denominator shown.

6 QUESTIONS

1 $\frac{1}{2} \rightarrow$ twelfths

4 $\frac{2}{5} \rightarrow$ tenths

2 $\frac{2}{3} \rightarrow$ twelfths

5 $\frac{5}{6} \rightarrow$ twelfths

3 $\frac{3}{4} \rightarrow$ twelfths

6 $\frac{3}{8} \rightarrow$ twenty-fourths



Level 2 · Building up

Medium · 4 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Order from smallest to largest.

a Order $\frac{4}{5}, \frac{3}{10}, \frac{2}{5}, \frac{7}{10}$ from smallest to largest.

Your teacher works this out. Copy every step.

b Order $\frac{1}{2}, \frac{1}{3}, \frac{7}{12}, \frac{11}{12}$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Order from smallest to largest.

4 QUESTIONS

REMEMBER Different denominators: find the LCM of the denominators.

1 $\frac{4}{5}, \frac{3}{10}, \frac{2}{5}, \frac{7}{10}$

3 $\frac{5}{8}, \frac{1}{2}, \frac{3}{4}, \frac{3}{8}$

2 $\frac{1}{2}, \frac{1}{3}, \frac{7}{12}, \frac{11}{12}$

4 $\frac{2}{3}, \frac{4}{9}, \frac{5}{6}, \frac{1}{2}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Order from smallest to largest.

a Order $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{6}$, $\frac{7}{12}$.

b Order $1\frac{1}{3}$, $\frac{5}{4}$, $\frac{11}{8}$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Order from smallest to largest.

4 QUESTIONS

REMEMBER Rewrite each fraction with that denominator.

1 $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{6}$, $\frac{7}{12}$

3 $1\frac{1}{3}$, $\frac{5}{4}$, $\frac{11}{8}$, $1\frac{1}{6}$

2 $\frac{3}{5}$, $\frac{1}{2}$, $\frac{7}{10}$, $\frac{4}{15}$

4 Find a fraction between $\frac{1}{3}$ and $\frac{1}{2}$.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Which is larger: $\frac{4}{9}$ or $\frac{7}{9}$?

Level 2

Which is larger: $\frac{3}{4}$ or $\frac{4}{5}$?

Level 3

Order $\frac{5}{6}$, $\frac{2}{3}$, $\frac{3}{4}$.

Extension · Stretch yourself**Worked example**

Order $\frac{5}{7}$, $\frac{2}{3}$, $\frac{7}{10}$ from smallest to largest.

1. Common denominator 210

2. $\frac{150}{210}$, $\frac{140}{210}$, $\frac{147}{210}$

Answer: $\frac{2}{3}$, $\frac{7}{10}$, $\frac{5}{7}$

YOU DO

On your own: read the worked example, then try these.

E1 Order $\frac{4}{9}$, $\frac{3}{7}$, $\frac{1}{2}$.

E2 Find a fraction exactly halfway between $\frac{1}{4}$ and $\frac{1}{3}$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Same denominator: compare the numerators.
 2. Different denominators: find the LCM of the denominators.
 3. Rewrite each fraction with that denominator.
 4. Order the numerators.

Level 1

$\frac{3}{8}$ or $\frac{5}{8}$

1. Same denominator

Answer: $\frac{5}{8}$

Level 2

$\frac{2}{3}$ or $\frac{3}{5}$

1. $\frac{10}{15}$ vs $\frac{9}{15}$

Answer: $\frac{2}{3}$

Level 3

$\frac{1}{2}$, $\frac{1}{3}$, $\frac{7}{12}$, $\frac{11}{12}$

1. Twelfths: 6, 4, 7, 11

Answer: $\frac{1}{3}$, $\frac{1}{2}$, $\frac{7}{12}$, $\frac{11}{12}$

Extension

$\frac{5}{7}$, $\frac{2}{3}$, $\frac{7}{10}$

1. Denominator 210

Answer: $\frac{2}{3}$, $\frac{7}{10}$, $\frac{5}{7}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.03

Adding and subtracting fractions

Date _____

Learning intentions

We are learning to:

- add and subtract fractions

Success criteria

I can:

- ☐ add and subtract fractions with the same denominator
- ☐ use a common denominator when the denominators are different
- ☐ simplify the answer

Key terms Write the definition as your teacher explains it.

Numerator

Denominator

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. The denominators are the same.

a $\frac{3}{4} + \frac{1}{4}$

b $\frac{5}{7} - \frac{2}{7}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate. Simplify if you can.

12 QUESTIONS

REMEMBER Same denominator: add or subtract the numerators.

1 $\frac{3}{4} + \frac{1}{4}$

2 $\frac{2}{9} + \frac{5}{9}$

3 $\frac{7}{10} - \frac{3}{10}$

4 $\frac{5}{8} - \frac{1}{8}$

5 $\frac{4}{5} + \frac{3}{5}$

6 $\frac{11}{12} - \frac{5}{12}$

Questions 7–12: evaluate.

7 $\frac{1}{5} + \frac{2}{5}$

8 $\frac{7}{9} - \frac{4}{9}$

9 $\frac{1}{2} + \frac{1}{4}$

10 $\frac{2}{3} - \frac{1}{6}$

11 $\frac{1}{3} + \frac{1}{4}$

12 $\frac{3}{4} - \frac{2}{3}$

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $\frac{1}{2} + \frac{1}{4}$

4 $\frac{5}{10} - \frac{1}{3}$

2 $\frac{2}{3} - \frac{1}{6}$

5 $\frac{1}{2} - \frac{3}{8}$

3 $\frac{2}{5} + \frac{3}{10}$

6 $\frac{3}{4} + \frac{1}{8}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Use a common denominator.

a $\frac{2}{5} + \frac{3}{10}$

Your teacher works this out. Copy every step.

b $\frac{3}{4} - \frac{1}{5}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate. Use a common denominator.

6 QUESTIONS

REMEMBER Different: find the lowest common denominator.

1 $\frac{1}{2} + \frac{1}{3}$

3 $\frac{2}{3} + \frac{1}{4}$

5 $\frac{5}{6} + \frac{3}{4}$

2 $\frac{3}{4} - \frac{1}{5}$

4 $\frac{4}{5} - \frac{1}{2}$

6 $\frac{13}{18} - \frac{3}{8}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Use the lowest common denominator.

a $\frac{1}{2} + \frac{1}{3}$

b $\frac{5}{6} + \frac{3}{4}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Rewrite, then add or subtract.

1 $\frac{5}{6} + \frac{4}{5}$

3 $\frac{7}{8} - \frac{1}{3} - \frac{1}{6}$

2 $\frac{1}{2} + \frac{1}{3} + \frac{1}{4}$

4 Tom ate $\frac{1}{3}$ of a pizza and Sue ate $\frac{2}{5}$. What fraction is left?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{2}{7} + \frac{3}{7}$

Level 2

$\frac{1}{2} - \frac{1}{6}$

Level 3

$\frac{2}{3} + \frac{3}{5}$

Extension · Stretch yourself**Worked example**

Find the missing fraction: $\frac{3}{8} + ? = \frac{5}{6}$

1. $? = \frac{5}{6} - \frac{3}{8}$

2. $= \frac{20}{24} - \frac{9}{24}$

Answer: $\frac{11}{24}$

YOU DO

On your own: read the worked example, then try these.

E1 Find the missing fraction: $? - \frac{2}{5} = \frac{1}{3}$

E2 Write $\frac{7}{12}$ as the sum of two different unit fractions (numerator 1).

Lesson summary

Missed the lesson? Start here.

Remember

1. Same denominator: add or subtract the numerators.
3. Rewrite, then add or subtract.

2. Different: find the lowest common denominator.

4. Simplify the answer.

Level 1

$$\frac{3}{4} + \frac{1}{4}$$

1. $\frac{4}{4}$

Answer: 1

Level 2

$$\frac{2}{5} + \frac{3}{10}$$

1. $\frac{4}{10} + \frac{3}{10}$

Answer: $\frac{7}{10}$

Level 3

$$\frac{3}{4} - \frac{1}{5}$$

1. $\frac{15}{20} - \frac{4}{20}$

Answer: $\frac{11}{20}$

Extension

$$\frac{3}{8} + ? = \frac{5}{6}$$

1. $\frac{20}{24} - \frac{9}{24}$

Answer: $\frac{11}{24}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.04

Adding and subtracting mixed numerals

Date _____

Learning intentions

We are learning to:

- add and subtract mixed numerals

Success criteria

I can:

- ☐ add the whole numbers, then the fractions
- ☐ change mixed numerals to improper fractions when subtracting needs it
- ☐ write the answer as a simplified mixed numeral

Key terms Write the definition as your teacher explains it.

Mixed numeral

Whole number

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate. Add the whole numbers, then the fractions.

a $2\frac{1}{5} + 6\frac{1}{2}$

b $3\frac{3}{4} + 1\frac{1}{2}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate.

12 QUESTIONS

REMEMBER Adding: add wholes, then add fractions.

1 $2\frac{1}{5} + 1\frac{2}{5}$

2 $3\frac{1}{4} + 2\frac{1}{4}$

3 $4\frac{5}{7} - 1\frac{2}{7}$

4 $5\frac{1}{2} + 3$

5 $6\frac{7}{9} - 2\frac{4}{9}$

6 $1\frac{1}{3} + 1\frac{2}{3}$

Questions 7–12: evaluate.

7 $2\frac{1}{3} + 1\frac{1}{3}$

8 $4\frac{3}{5} - 2\frac{1}{5}$

9 $3 - \frac{1}{4}$

10 $1\frac{1}{2} + 1\frac{1}{2}$

11 $2\frac{3}{4} + \frac{1}{2}$

12 $5\frac{1}{6} - 1\frac{5}{6}$

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $10 - \frac{1}{3}$

4 $4 - 1\frac{2}{5}$

2 $5 - \frac{3}{4}$

5 $3\frac{5}{6} + \frac{1}{6}$

3 $2\frac{3}{4} + 1\frac{3}{4}$

6 $2\frac{1}{8} - \frac{3}{8}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate.

a $10 - \frac{1}{3}$

Your teacher works this out. Copy every step.

b $3\frac{2}{3} - 2\frac{1}{6}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Carry if the fraction is more than 1.

1 $2\frac{1}{5} + 6\frac{1}{2}$

3 $3\frac{3}{4} + 1\frac{1}{2}$

5 $1\frac{2}{3} + 2\frac{3}{4}$

2 $3\frac{2}{3} - 2\frac{1}{6}$

4 $4\frac{3}{4} - 2\frac{1}{3}$

6 $3\frac{5}{6} - 1\frac{1}{4}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions.

a $5\frac{1}{4} - 3\frac{2}{3}$

b $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Subtracting: change to improper fractions if needed.

1 $5\frac{1}{4} - 3\frac{2}{3}$

3 $2\frac{1}{2} + 1\frac{1}{3} + 2\frac{3}{4}$

2 $3\frac{1}{2} - 1\frac{4}{5}$

4 A plank is $3\frac{1}{2}$ m long. $1\frac{3}{4}$ m is cut off. How long is the rest?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$2\frac{1}{3} + 4\frac{1}{3}$

Level 2

$6 - \frac{2}{5}$

Level 3

$4\frac{1}{3} - 2\frac{3}{4}$

Extension · Stretch yourself**Worked example**Evaluate $2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$.

1. Improper: $\frac{11}{4} - \frac{8}{5} + \frac{3}{2}$
2. Denominator 20: $\frac{55}{20} - \frac{32}{20} + \frac{30}{20}$
3. $\frac{53}{20}$

Answer: $2\frac{13}{20}$ **YOU DO**

On your own: read the worked example, then try these.

E1 Evaluate $5\frac{1}{3} - 2\frac{1}{2} - 1\frac{3}{4}$.

E2 Two mixed numerals add to $7\frac{1}{6}$. One is $3\frac{2}{3}$. Find the other.

Lesson summary

Missed the lesson? Start here.

Remember

1. Adding: add wholes, then add fractions.
2. Carry if the fraction is more than 1.
3. Subtracting: change to improper fractions if needed.
4. Write the answer as a mixed numeral.

Level 1

$2\frac{1}{5} + 6\frac{1}{2}$

1. $8 + \frac{7}{10}$

Answer: $8\frac{7}{10}$ **Level 2**

$10 - \frac{1}{3}$

1. $9\frac{3}{3} - \frac{1}{3}$

Answer: $9\frac{2}{3}$ **Level 3**

$5\frac{1}{4} - 3\frac{2}{3}$

1. $\frac{63}{12} - \frac{44}{12}$

Answer: $1\frac{7}{12}$ **Extension**

$2\frac{3}{4} - 1\frac{3}{5} + 1\frac{1}{2}$

1. $\frac{53}{20}$

Answer: $2\frac{13}{20}$

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LESSON
4.05

Fraction of a quantity

Date _____

Learning intentions

We are learning to:

- find a fraction of a quantity, including with units

Success criteria

I can:

- ☐ divide by the denominator, then multiply by the numerator
- ☐ convert units before finding the fraction
- ☐ give the answer in the unit asked for

Key terms Write the definition as your teacher explains it.**Quantity****Unit**

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Evaluate.

a $\frac{1}{4}$ of 28

b $\frac{3}{7}$ of 21

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate.

12 QUESTIONS

REMEMBER “of” means multiply.

1 $\frac{1}{4}$ of 28

2 $\frac{1}{10}$ of 35

3 $\frac{2}{5}$ of 20

4 $\frac{4}{9}$ of 27

5 $\frac{2}{3}$ of 30

6 $\frac{3}{7}$ of 21

Questions 7–12: evaluate.

7 $\frac{1}{3}$ of 18

8 $\frac{1}{5}$ of 40

9 $\frac{3}{4}$ of 20

10 $\frac{2}{3}$ of 12

11 $\frac{5}{8}$ of 32

12 $\frac{3}{10}$ of 50

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $\frac{3}{8}$ of 64

4 $\frac{3}{4}$ of 1000

2 $\frac{5}{6}$ of 42

5 $\frac{2}{9}$ of 54

3 $\frac{7}{10}$ of 150

6 $\frac{5}{12}$ of 96



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate in the unit given.

a $\frac{2}{3}$ of 1 day (in hours)

Your teacher works this out. Copy every step.

b $\frac{4}{5}$ of 2 km (in metres)

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate in the unit shown.

6 QUESTIONS

REMEMBER Divide by the denominator.

1 $\frac{2}{3}$ of 1 day (hours)

3 $\frac{3}{4}$ of 3 h (min)

5 $\frac{1}{6}$ of 5 min (s)

2 $\frac{4}{5}$ of 2 km (m)

4 $\frac{7}{10}$ of 15 cm (mm)

6 $\frac{5}{8}$ of 2 m (cm)

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate in the unit given.

a $\frac{3}{4}$ of 3 hours (in minutes)

b $\frac{1}{6}$ of 5 minutes (in seconds)

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Word problems.

4 QUESTIONS

REMEMBER Multiply by the numerator.

1 $\frac{3}{5}$ of a class of 30 are girls. How many boys are there?

3 $\frac{2}{3}$ of a number is 24. Find the number.

2 A phone costs \$840. Mia has saved $\frac{5}{7}$ of the price. How much more does she need?

4 Which is more: $\frac{3}{4}$ of an hour or $\frac{7}{10}$ of an hour? By how many minutes?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{1}{5}$ of 45

Level 2

$\frac{5}{6}$ of 72

Level 3

$\frac{3}{4}$ of 2 hours (in minutes)

Extension · Stretch yourself**Worked example**

After spending $\frac{1}{4}$ of his money and then $\frac{1}{3}$ of what was left, Ali has \$30. How much did he start with?

1. After $\frac{1}{4}$: $\frac{3}{4}$ left
2. Then $\frac{2}{3}$ of $\frac{3}{4} = \frac{1}{2}$ of the money is left
3. $\frac{1}{2} = \$30$

Answer: \$60**YOU DO**

On your own: read the worked example, then try these.

- E1** $\frac{2}{5}$ of a number is 18. What is $\frac{3}{4}$ of the number?

- E2** A tank is $\frac{3}{8}$ full. After adding 60 L it is $\frac{3}{4}$ full. How much does it hold?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. “of” means multiply.
 3. Multiply by the numerator.

2. Divide by the denominator.
4. Change units first if needed.

Level 1

$\frac{3}{7}$ of 21

1. $21 \div 7 = 3$
2. 3×3

Answer: 9**Level 2**

$\frac{4}{5}$ of 2 km in m

1. 2 km = 2000 m
2. $2000 \div 5 \times 4$

Answer: 1600 m**Level 3**

$\frac{3}{4}$ of 3 h in min

1. $180 \div 4 \times 3$

Answer: 135 min**Extension**

Ali's money

1. $\frac{1}{2}$ left is \$30

Answer: \$60

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LESSON
4.06

Multiplying fractions

Date _____

Learning intentions

We are learning to:

- multiply fractions and mixed numerals

Success criteria

I can:

- ☐ multiply numerators and multiply denominators
- ☐ cancel common factors first
- ☐ change mixed numerals to improper fractions first

Key terms Write the definition as your teacher explains it.

Product

Cancel

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate.

a $\frac{2}{3} \times \frac{1}{5}$

b $10 \times \frac{3}{5}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate.

12 QUESTIONS

REMEMBER Change mixed numerals to improper fractions.

1 $\frac{2}{3} \times \frac{1}{5}$

2 $\frac{1}{3} \times \frac{1}{7}$

3 $\frac{3}{4} \times \frac{1}{2}$

4 $\frac{2}{5} \times \frac{2}{3}$

5 $10 \times \frac{3}{5}$

6 $\frac{5}{6} \times 12$

Questions 7–12: evaluate.

7 $\frac{1}{2} \times \frac{1}{3}$

8 $\frac{2}{5} \times \frac{3}{4}$

9 $\frac{3}{8} \times 4$

10 $\frac{5}{6} \times \frac{3}{10}$

11 $6 \times \frac{2}{3}$

12 $\frac{4}{9} \times \frac{3}{8}$

YOU DO

On your own: **Easy** Set B. Evaluate. Cancel first.

6 QUESTIONS

1 $\frac{4}{7} \times \frac{5}{8}$

4 $\frac{9}{16} \times \frac{2}{45}$

2 $\frac{3}{10} \times \frac{5}{9}$

5 $\frac{4}{15} \times \frac{5}{8}$

3 $\frac{6}{7} \times \frac{7}{12}$

6 $\frac{2}{3} \times \frac{9}{10}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Cancel first.

a $\frac{4}{7} \times \frac{5}{8}$

Your teacher works this out. Copy every step.

b $\frac{9}{16} \times \frac{2}{45}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Cancel common factors (top with bottom).

1 $1\frac{1}{2} \times \frac{1}{2}$

3 $1\frac{1}{2} \times 1\frac{1}{2}$

5 $\frac{3}{4} \times 2\frac{1}{3}$

2 $2\frac{1}{5} \times \frac{1}{2}$

4 $2\frac{2}{3} \times \frac{1}{2}$

6 $3\frac{3}{5} \times \frac{3}{8}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions first.

a $1\frac{1}{2} \times \frac{1}{2}$

b $2\frac{1}{3} \times 4\frac{2}{3}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Multiply tops, multiply bottoms.

1 $2\frac{1}{3} \times 4\frac{2}{3}$

3 $2\frac{1}{2} \times 1\frac{2}{5} \times \frac{1}{3}$

2 $3\frac{2}{4} \times \frac{2}{3}$

4 A recipe needs $2\frac{3}{4}$ cups of flour. How much for $\frac{2}{3}$ of the recipe?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{3}{5} \times \frac{1}{4}$

Level 2

$\frac{5}{9} \times \frac{3}{10}$

Level 3

$1\frac{1}{4} \times 2\frac{2}{3}$

Extension · Stretch yourself**Worked example**Evaluate $\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \frac{5}{6}$.

1. Cancel: 3, 4 and 5 cancel
2. Leaves $\frac{2}{6}$

Answer: $\frac{1}{3}$ **YOU DO**

On your own: read the worked example, then try these.

E1 Evaluate $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{9}{10}$.**E2** A rectangle is $3\frac{1}{2}$ m by $2\frac{2}{3}$ m. Find its area.**Lesson summary**

Missed the lesson? Start here.

- Remember**
1. Change mixed numerals to improper fractions.
 2. Cancel common factors (top with bottom).
 3. Multiply tops, multiply bottoms.
 4. Write as a simplified mixed numeral.

Level 1

$$\frac{2}{3} \times \frac{1}{5}$$

1. 2×1 and 3×5

Answer: $\frac{2}{15}$ **Level 2**

$$\frac{4}{7} \times \frac{5}{8}$$

1. Cancel 4 and 8

Answer: $\frac{5}{14}$ **Level 3**

$$2\frac{1}{3} \times 4\frac{2}{3}$$

$$1. \frac{7}{3} \times \frac{14}{3}$$

Answer: $10\frac{8}{9}$ **Extension**

Telescoping product

1. Cancel

Answer: $\frac{1}{3}$

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LESSON
4.07

Dividing fractions

Date _____

Learning intentions

We are learning to:

- divide fractions and mixed numerals

Success criteria

I can:

- ☐ find the **reciprocal** of a fraction
- ☐ divide by multiplying by the reciprocal
- ☐ change mixed numerals to improper fractions first

Key terms Write the definition as your teacher explains it.**Reciprocal****Quotient**

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Write the reciprocal, then divide.**a** Write the reciprocal of $\frac{3}{4}$, 5 and $2\frac{1}{2}$.**b** $10 \div \frac{1}{5}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write the reciprocal.

12 QUESTIONS

REMEMBER Change mixed numerals to improper fractions.

1 $\frac{3}{4}$

2 $\frac{2}{5}$

3 7

4 $\frac{1}{9}$

5 $\frac{8}{3}$

6 $1\frac{1}{2}$

Questions 7–12: evaluate.

7 $6 \div \frac{1}{2}$

8 $\frac{1}{3} \div 2$

9 $\frac{3}{4} \div \frac{1}{4}$

10 $\frac{2}{5} \div \frac{4}{5}$

11 $\frac{5}{6} \div \frac{1}{3}$

12 $4 \div \frac{2}{3}$

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $10 \div \frac{1}{5}$

4 $6 \div \frac{2}{3}$

2 $\frac{1}{2} \div 3$

5 $\frac{4}{5} \div 2$

3 $\frac{1}{2} \div \frac{1}{3}$

6 $3 \div \frac{1}{4}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate. Multiply by the reciprocal.

a $\frac{1}{2} \div 3$

Your teacher works this out. Copy every step.

b $\frac{3}{4} \div \frac{3}{8}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Keep the first fraction.

1 $\frac{3}{4} \div \frac{3}{8}$

3 $\frac{4}{7} \div \frac{5}{8}$

5 $\frac{2}{3} \div \frac{5}{6}$

2 $\frac{3}{5} \div \frac{1}{10}$

4 $\frac{17}{20} \div \frac{3}{5}$

6 $\frac{5}{9} \div \frac{2}{3}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate. Change to improper fractions first.

a $2\frac{1}{4} \div 1\frac{1}{8}$

b $1\frac{7}{8} \div 1\frac{1}{4}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Change $\div$ to $\times$ and flip the second fraction.

1 $1\frac{1}{2} \div \frac{1}{2}$

3 $1\frac{7}{8} \div 1\frac{1}{4}$

2 $2\frac{1}{4} \div 1\frac{1}{8}$

4 How many $\frac{3}{4}$ m pieces can be cut from a $7\frac{1}{2}$ m rope?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write the reciprocal of $\frac{5}{8}$.

Level 2

$\frac{2}{3} \div \frac{1}{6}$

Level 3

$3\frac{1}{2} \div 1\frac{3}{4}$

Extension · Stretch yourself**Worked example**

A jug holds $2\frac{1}{4}$ L. How many $\frac{3}{10}$ L glasses can be filled?

$$1. \frac{9}{4} \div \frac{3}{10} = \frac{9}{4} \times \frac{10}{3}$$

$$2. = \frac{90}{12} = 7\frac{1}{2}$$

Answer: 7 full glasses

YOU DO

On your own: read the worked example, then try these.

E1 $\frac{2}{3}$ of a number is $\frac{4}{9}$. Find the number.

E2 Evaluate $(\frac{1}{2} + \frac{1}{3}) \div (\frac{1}{2} - \frac{1}{3})$.

Lesson summary

Missed the lesson? Start here.

Remember 1. Change mixed numerals to improper fractions. 2. Keep the first fraction.
3. Change $\div$ to $\times$ and flip the second fraction. 4. Multiply and simplify.

Level 1

$$10 \div \frac{1}{5}$$

$$1. 10 \times 5$$

Answer: 50

Level 2

$$\frac{3}{4} \div \frac{3}{8}$$

$$1. \frac{3}{4} \times \frac{8}{3}$$

Answer: 2

Level 3

$$2\frac{1}{4} \div 1\frac{1}{8}$$

$$1. \frac{9}{4} \times \frac{8}{9}$$

Answer: 2

Extension

Glasses from a jug

$$1. 7\frac{1}{2}$$

Answer: 7 glasses

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.08

Fraction problems

Date _____

Learning intentions

We are learning to:

- solve problems involving fractions

Success criteria

I can:

- ☐ decide which operation the problem needs
- ☐ find the “rest” by subtracting fractions from 1
- ☐ check the answer makes sense

Key terms Write the definition as your teacher explains it.

Remainder

Total

Notes

Level 1 · Getting started

Easy · 15 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Read the problem, then answer.

- a** A punch has 2 L ginger ale, $1\frac{1}{2}$ L orange juice, $2\frac{3}{4}$ L lemonade and $\frac{1}{3}$ L raspberry juice. How many litres of punch?

- b** How much more lemonade than orange juice is in the punch?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Solve.

6 QUESTIONS

REMEMBER Read the problem twice. Underline the numbers.

- ① Sam ate $\frac{1}{4}$ of a cake and Ben ate $\frac{3}{8}$. How much did they eat together?

- ③ $\frac{3}{5}$ of 40 students walk to school. How many walk?

- ⑤ A pizza has 8 slices. Lee eats 3. What fraction is left?

- ② A tank is $\frac{5}{6}$ full. $\frac{1}{3}$ is used. How full is it now?

- ④ What fraction of an hour is 20 minutes?

- ⑥ How many $\frac{1}{4}$ -hour lessons fit in 3 hours?

YOU DO On your own: **Easy** Set A2. Quick drill: what fraction is left?

6 QUESTIONS

- ① $\frac{1}{4}$ is eaten

- ② $\frac{2}{5}$ is spent

- ③ $\frac{1}{3}$ and $\frac{1}{3}$ are used

- ④ $\frac{1}{2}$ and $\frac{1}{4}$ are sold

- ⑤ $\frac{3}{8}$ is poured out

- ⑥ $\frac{1}{6}$ and $\frac{1}{3}$ are given away

YOU DO On your own: **Easy** Set B. Solve.

3 QUESTIONS

- ① A ribbon is $4\frac{1}{2}$ m. $1\frac{2}{3}$ m is cut off. How much is left?

- ② A bag of flour weighs $\frac{3}{4}$ kg. What do 6 bags weigh?

- ③ $\frac{1}{3}$ of the class have a dog and $\frac{1}{4}$ have a cat. What fraction have neither (no one has both)?

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Read the problem, then answer.

- a** A farmer has 24 goats. $\frac{1}{4}$ are black and $\frac{2}{3}$ are white. What fraction are neither?

Your teacher works this out. Copy every step.

- b** The rest of the goats are split equally between brown and spotted. How many are spotted?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Solve.

6 QUESTIONS

REMEMBER “Of” → multiply. “Altogether” → add.

- 1** A punch uses $1\frac{1}{2}$ L juice, $2\frac{3}{4}$ L soda and $\frac{1}{3}$ L syrup. Find the total.

- 3** A 60 L tank is $\frac{3}{5}$ full. How many litres to fill it?

- 5** How many $\frac{2}{3}$ -cup serves are in 8 cups?

- 2** 24 goats: $\frac{1}{4}$ black, $\frac{2}{3}$ white, the rest split equally brown and spotted. How many spotted?

- 4** Mia spends $\frac{1}{3}$ of \$90, then $\frac{1}{2}$ of what is left. How much is left?

- 6** A car uses $\frac{3}{4}$ of a tank for 360 km. How far on a full tank?

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Read the problem, then answer.

- a** Jo reads $\frac{2}{5}$ of a book on Monday and $\frac{1}{3}$ on Tuesday. What fraction is left?

Your teacher works this out. Copy every step.

- b** The book has 240 pages. How many pages are left?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Harder problems.

4 QUESTIONS

REMEMBER “Left” or “the rest” → subtract from 1 (the whole).

- 1** $\frac{3}{8}$ of a school's 480 students catch a bus and $\frac{1}{3}$ of the rest walk. How many walk?

- 3** After giving away $\frac{2}{7}$ of her cards, Zoe has 45 left. How many did she start with?

- 2** A rope is cut into pieces of $\frac{2}{5}$ m. There are 12 pieces and $\frac{1}{10}$ m left over. How long was the rope?

- 4** Is $\frac{2}{3}$ of $\frac{3}{4}$ of 80 bigger than $\frac{1}{2}$ of 80? Explain.



Extension · Stretch yourself

Finished Set D? Stretch yourself here.



Worked example

A cyclist rides $\frac{1}{3}$ of a trip on day 1 and $\frac{1}{4}$ of the trip on day 2. She has 50 km left. How long is the trip?

1. $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$
2. $\frac{5}{12}$ of the trip is 50 km
3. $\frac{1}{12}$ is 10 km

Answer: 120 km

YOU DO

On your own: read the worked example, then try these.

- E1 $\frac{1}{4}$ of a class plays soccer, $\frac{1}{3}$ plays netball and the other 10 play neither. How many in the class?

- E2 A bottle is $\frac{2}{3}$ full. After pouring out 150 mL it is $\frac{1}{6}$ full. How much does it hold?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is $\frac{2}{5} + \frac{1}{3}$ of a whole?

Level 2

 $\frac{3}{4}$ of 36 students passed. How many failed?

Level 3

After spending $\frac{3}{5}$ of his money, Kai has \$28. How much did he start with?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Read the problem twice. Underline the numbers.
 2. “Of” → multiply. “Altogether” → add.
 3. “Left” or “the rest” → subtract from 1 (the whole).
 4. Answer the question with units.

Level 1

Punch total

1. $2 + 1\frac{1}{2} + 2\frac{3}{4} + \frac{1}{3}$

Answer: $6\frac{7}{12}$ L

Level 2

Goats: neither black nor white

1. $1 - \frac{1}{4} - \frac{2}{3}$

Answer: $\frac{1}{12}$

Level 3

Spotted goats

1. $\frac{1}{12}$ of 24 = 2
2. $2 \div 2$

Answer: 1 goat

Extension

Cyclist

1. $\frac{5}{12} = 50$ km

Answer: 120 km

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
4.09

Fractions on a calculator

Date _____

Learning intentions

We are learning to:

- use a calculator to work with fractions

Success criteria

I can:

- ☐ enter fractions and mixed numerals with the fraction key
- ☐ convert between improper fractions, mixed numerals and decimals
- ☐ check calculator answers with an estimate

Key terms Write the definition as your teacher explains it.

Fraction key

S ⇌ D key

Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the fraction key on your calculator.

a Enter $\frac{3}{4} + \frac{1}{6}$.

b Enter $3\frac{1}{2} \times \frac{2}{5}$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Use a calculator.

10 QUESTIONS

REMEMBER Use the fraction key to enter a fraction.

1 $\frac{3}{4} + \frac{1}{6}$

2 $\frac{5}{6} + \frac{4}{5}$

3 $\frac{7}{9} - \frac{1}{4}$

4 $\frac{13}{18} - \frac{3}{8}$

5 $2\frac{3}{4} + 1\frac{5}{6}$

6 $5\frac{1}{2} - 2\frac{2}{3}$

Questions 7–10: use your calculator.

7 $\frac{5}{12} + \frac{7}{18}$

8 $\frac{11}{15} - \frac{3}{10}$

9 $\frac{7}{12} \times \frac{9}{14}$

10 $\frac{5}{9} \div \frac{10}{27}$

YOU DO

On your own: **Easy** Set B. Use a calculator.

6 QUESTIONS

1 $\frac{9}{16} \times \frac{2}{45}$

4 $2\frac{3}{4} \div 1\frac{1}{5}$

2 $\frac{17}{20} \div \frac{3}{5}$

5 $\frac{5}{8}$ of 360

3 $2\frac{1}{3} \times 4\frac{2}{3}$

6 $\frac{7}{12} \times 48$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use your calculator to convert and evaluate.

- a** Convert $\frac{17}{5}$ to a mixed numeral and to a decimal.

Your teacher works this out. Copy every step.

- b** Evaluate $\frac{13}{18} - \frac{3}{8}$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Convert with a calculator.

6 QUESTIONS

REMEMBER Use SHIFT for a mixed numeral.

1 $\frac{17}{5}$ to a mixed numeral

3 $2\frac{3}{8}$ to an improper fraction

5 0.625 to a fraction

2 $\frac{45}{8}$ to a mixed numeral

4 $\frac{7}{20}$ to a decimal

6 $\frac{5}{16}$ to a decimal

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use your calculator. Use brackets.

a Evaluate $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$.

Your teacher works this out. Copy every step.

b Evaluate $2\frac{1}{4}^2$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use a calculator.

4 QUESTIONS

REMEMBER S $\leftrightarrow$ D switches between fraction and decimal.

1 $4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$

3 $(\frac{3}{5} + \frac{1}{4}) \times 2\frac{2}{3}$

2 $2\frac{1}{4}^2 + \frac{1}{2}$

4 Estimate $3\frac{7}{8} \times 2\frac{1}{10}$ first, then use a calculator.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

$\frac{2}{7} + \frac{3}{5}$

Level 2

$3\frac{4}{5} \times \frac{5}{6}$

Level 3

$2\frac{5}{6} \div \frac{2}{3} - \frac{1}{4}$

Extension · Stretch yourself**Worked example**

Find the fraction halfway between $\frac{5}{9}$ and $\frac{7}{11}$.

1. Add and divide by 2

2. $(\frac{5}{9} + \frac{7}{11}) \div 2$

Answer: $\frac{59}{99}$

YOU DO

On your own: read the worked example, then try these.

E1 Evaluate $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}$. What do you notice?

E2 Which is closer to 1: $\frac{11}{13}$ or $\frac{13}{15}$?

Lesson summary

Missed the lesson? Start here.

Remember

1. Use the fraction key to enter a fraction.
2. Use SHIFT for a mixed numeral.
3. S $\leftrightarrow$ D switches between fraction and decimal.
4. Estimate first to check the answer.

Level 1

$$\frac{3}{4} + \frac{1}{6}$$

1. Enter with the fraction key

Answer: $\frac{11}{12}$

Level 2

$$\frac{17}{5}$$

1. Convert to mixed

Answer: $3\frac{2}{5} = 3.4$

Level 3

$$4\frac{2}{3} \div \frac{1}{3} - 1\frac{3}{4}$$

1. Use brackets if unsure

Answer: $12\frac{1}{4}$

Extension

Halfway

1. Add and halve

Answer: $\frac{59}{99}$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.10

Percentages, fractions and decimals

Date _____

Learning intentions

We are learning to:

- convert between percentages, fractions and decimals

Success criteria

I can:

- ☐ write a percentage as a fraction out of 100 and simplify
- ☐ write a percentage as a decimal by dividing by 100
- ☐ write a fraction or decimal as a percentage by multiplying by 100%

Key terms Write the definition as your teacher explains it.

Percentage

Per cent

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write as a fraction and a decimal.

a Write 13% as a fraction and a decimal.

Your teacher works this out. Copy every step.

b Write 8% as a fraction and a decimal.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write each percentage as a fraction in simplest form.

12 QUESTIONS

REMEMBER % → fraction: write over 100, simplify.

1 13%

2 8%

3 50%

4 75%

5 35%

6 120%

Questions 7–12: complete the conversion.

7 25% = ? (fraction)

8 0.6 = ?%

9 $\frac{1}{5}$ = ?%

10 40% = ? (decimal)

11 $\frac{3}{4}$ = ?%

12 0.07 = ?%

YOU DO

On your own: **Easy** Set B. Write each percentage as a decimal.

6 QUESTIONS

1 13%

4 150%

2 8%

5 2.5%

3 45%

6 $7\frac{1}{2}\%$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write as a fraction and a decimal, then as a percentage.

a Write $7\frac{1}{2}\%$ as a fraction and a decimal.

Your teacher works this out. Copy every step.

b Write $\frac{7}{10}$ as a percentage.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Write each as a percentage.

6 QUESTIONS

REMEMBER % → decimal: ÷ 100 (move 2 places left).

1 $\frac{7}{10}$

4 0.6

2 $\frac{23}{25}$

5 0.045

3 $\frac{3}{4}$

6 1.3

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write as a percentage.

a Write $\frac{23}{25}$ as a percentage.

Your teacher works this out. Copy every step.

b Write 0.045 as a percentage.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Complete the conversions.

4 QUESTIONS

REMEMBER Fraction or decimal $\rightarrow$ %: $\times 100\%$.

1 Write $\frac{3}{8}$ as a decimal and a percentage.

3 Write $12\frac{1}{2}\%$ as a fraction.

2 Write $\frac{1}{3}$ as a percentage.

4 Order from smallest to largest: 0.4, 38%, $\frac{2}{5}$, $\frac{3}{8}$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 25% as a fraction.

Level 2

Write 0.07 as a percentage.

Level 3

Write $\frac{3}{16}$ as a percentage.

Extension · Stretch yourself**Worked example**

Write $0.\overline{3}$ ($0.333\dots$) as a percentage and as a fraction.

1. $0.333\dots \times 100\% = 33.3\dots\%$

2. $= 33\frac{1}{3}\%$

3. As a fraction: $\frac{1}{3}$

Answer: $33\frac{1}{3}\% = \frac{1}{3}$

YOU DO

On your own: read the worked example, then try these.

E1 Write $\frac{2}{3}$ as a percentage.

E2 Write $12\frac{1}{3}\%$ as a fraction in simplest form.

Lesson summary

Missed the lesson? Start here.

Remember 1. % $\rightarrow$ fraction: write over 100, simplify.
3. Fraction or decimal $\rightarrow$ %: $\times 100\%$.

2. % $\rightarrow$ decimal: $\div 100$ (move 2 places left).
4. Learn: $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{1}{5} = 20\%$, $\frac{1}{10} = 10\%$.

Level 1

13%

1. $\frac{13}{100}$

Answer: $\frac{13}{100} = 0.13$

Level 2

$7\frac{1}{2}\%$

1. $\frac{15}{2} \div 100 = \frac{15}{200}$

Answer: $\frac{3}{40} = 0.075$

Level 3

$\frac{23}{25}$

1. $\frac{92}{100}$

Answer: 92%

Extension

$\frac{1}{3}$

1. $\times 100\% = \frac{100}{3}\%$

Answer: $33\frac{1}{3}\%$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.11

Percentage of a quantity

Date _____

Learning intentions

We are learning to:

- find a percentage of a quantity

Success criteria

I can:

- ☐ change the percentage to a fraction or decimal and multiply
- ☐ use 10% and 1% to find percentages mentally
- ☐ solve problems with percentages

Key terms Write the definition as your teacher explains it.**Discount****Percentage**

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.**Example 1** Find mentally.**a** Find 10% of 350.

Your teacher works this out. Copy every step.

b Find 25% of 84.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find mentally.

12 QUESTIONS

REMEMBER Change % to a fraction or decimal.**1** 10% of 350

2 50% of 86

3 25% of 84

4 1% of 900

5 20% of 45

6 5% of 60

Questions 7–12: find mentally.

7 10% of 70

8 50% of 64

9 25% of 120

10 1% of 300

11 20% of 55

12 75% of 40

YOU DO On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 15% of \$60

4 75% of 36

2 7% of 1200

5 2% of 450

3 30% of 250

6 150% of 40

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Find the percentage of the quantity.

a Find 15% of \$60.

Your teacher works this out. Copy every step.

b Find 7% of 1200.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER Multiply by the quantity.

1 12.5% of 480 kg

3 8% of 2.5 km (m)

5 17% of 300

2 35% of \$180

4 45% of 3 h (min)

6 0.5% of \$2000

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Find, then solve the problem.

a Find 12.5% of 480 kg.

Your teacher works this out. Copy every step.

b A \$250 jacket is 30% off. Find the sale price.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Percentage problems.

4 QUESTIONS

REMEMBER 10%: ÷ 10. 1%: ÷ 100. Build others from these.

1 A \$250 jacket is 30% off. Find the sale price.

3 A \$40 meal has a 15% tip added. Find the total.

2 64% of 25 students caught a bus. How many did not?

4 Which is more: 30% of 70 or 70% of 30? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find 10% of 80.

Level 2

Find 35% of 200.

Level 3

A \$60 shirt is 15% off. Find the sale price.

 Extension · Stretch yourself Worked example

A price rises by 10%, then falls by 10%. Is it back to the original price?

1. Try $\$100: + 10\% = \110
2. $- 10\% \text{ of } \$110 = \$110 - \$11 = \99

Answer: No: it is 1% lower

YOU DO

On your own: read the worked example, then try these.

- E1 A \$80 item has 20% off, then another 10% off the new price. Find the final price.

- E2 Is 20% off then 10% off the same as 30% off? Explain.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Change % to a fraction or decimal.
 2. Multiply by the quantity.
 3. $10\%: \div 10$. $1\%: \div 100$. Build others from these.
 4. Discount: subtract from the price.

 Level 1

10% of 350

1. $350 \div 10$

Answer: 35

 Level 2

15% of \$60

1. $10\% = \$6$, $5\% = \$3$

Answer: \$9

 Level 3

30% off \$250

1. $30\% \text{ of } 250 = 75$
2. $250 - 75$

Answer: \$175

 Extension

+10% then -10%

1. $100 \rightarrow 110 \rightarrow 99$

Answer: No, 1% less

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
4.12

Expressing as a fraction or percentage

Date _____

Learning intentions

We are learning to:

- express one quantity as a fraction or percentage of another

Success criteria

I can:

- ☐ write the part over the whole
- ☐ convert both quantities to the same unit first
- ☐ multiply by 100% to get a percentage

Key terms Write the definition as your teacher explains it.

Part

Whole

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write as a fraction and a percentage.

- a** Write 51 out of 60 as a fraction and a percentage.

Your teacher works this out. Copy every step.

- b** Write 18 out of 24 as a percentage.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write the first quantity as a fraction of the second, in simplest form. **12 QUESTIONS**

REMEMBER Write **PART** ÷ **WHOLE** as a fraction.

- 1** 6 out of 10 **2** 15 out of 20 **3** 51 out of 60 **4** 8 out of 12

- 5** 40 out of 100 **6** 9 out of 45

Questions 7–12: write as a percentage.

- 7** 3 out of 10 **8** 9 out of 20 **9** 12 out of 25 **10** 17 out of 50

- 11** 30 out of 40 **12** 4 out of 5

YOU DO

On your own: **Easy** Set B. Write as a percentage.

6 QUESTIONS

1 7 out of 10

4 13 out of 20

2 18 out of 24

5 45 out of 50

3 51 out of 60

6 3 out of 8

 Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Change to the same units, then write as a fraction and a percentage.

a Write 300 m as a fraction of 1.5 km.

Your teacher works this out. Copy every step.

b Write 300 m as a percentage of 1.5 km.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Change to the same unit, then write as a fraction and a percentage.

6 QUESTIONS

REMEMBER Use the same units for both.

1 300 m of 1.5 km

Patient Information	
First Name	
Last Name	
Address	
City	
State	
Zip	
Phone	
Age	
Gender	
Occupation	
Referral Source	
Medical History	
Past Medical History	
Current Medical History	
Allergies	
Social History	
Family History	
Physical Examination	
Vital Signs	
Laboratory Tests	
Imaging Studies	
Diagnosis	
Treatment Plan	
Follow-up	

3 25 cm of 2 m

[illegible]

5 \$1.20 of \$4

② 45 min of 2 h

4 400 g of 5 kg

[illegible]

6 750 mL of 3 L

[illegible]

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write as a percentage, then compare.

a Write 45 min as a percentage of 2 hours.

Your teacher works this out. Copy every step.

b Mia scored 34/40 and Jake 42/50. Who did better?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Problems.

4 QUESTIONS

REMEMBER Simplify the fraction.

1 Mia scored 34/40 and Jake scored 42/50. Who did better?

3 A \$120 price dropped to \$90. What is the percentage decrease?

2 A team won 18 of 24 games. What percentage did they lose?

4 A class of 28 has 12 boys. What percentage are girls? (1 d.p.)

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 12 out of 16 as a fraction.

Level 2

Write 21 out of 25 as a percentage.

Level 3

Write 40 cm as a percentage of 2 m.

Extension · Stretch yourself**Worked example**

A shirt that cost \$40 now costs \$46. Find the percentage increase.

1. Increase = \$6
2. $\frac{6}{40} \times 100\%$

Answer: 15%

YOU DO

On your own: read the worked example, then try these.

- E1** A town grew from 2500 to 3100 people. Find the percentage increase.

- E2** A \$75 game is on sale for \$60. What percentage discount is this?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write **part ÷ whole** as a fraction.
 2. Use the same units for both.
 3. Simplify the fraction.
 4. Multiply by 100% for a percentage.

Level 1

51 out of 60

1. $\frac{51}{60}$
2. $\times 100\%$

Answer: 85%

Level 2

300 m of 1.5 km

1. 1.5 km = 1500 m
2. $\frac{300}{1500}$

Answer: $\frac{1}{5} = 20\%$

Level 3

45 min of 2 h

1. $\frac{45}{120}$

Answer: $\frac{3}{8} = 37.5\%$

Extension

\$40 → \$46

1. $\frac{6}{40} \times 100\%$

Answer: 15%

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